

Instructions:

1. This question paper contains 50 questions. All questions are compulsory.
2. The duration of the examination is 90 minutes.
3. Each question carries 2 marks. There is no negative marks for wrong answers.
4. Use only blue or black ball-point pen to mark your answers on the OMR Answer Sheet.
5. The use of calculators, mobile phones, smart watches, log tables, or any electronic gadgets is strictly prohibited.

Mathematics

1. A ladder 10 m long reaches a window 8 m high. How far is the foot of the ladder from the wall?
 (A) 2 m (B) 6 m
 (C) 4 m (D) 12 m
2. Rohan divides a pizza into equal slices. If he eats $\frac{2}{5}$ of the pizza and then gives $\frac{1}{3}$ of the remaining to his friend, what fraction of the whole pizza is left?
 (A) $\frac{2}{5}$ (B) $\frac{4}{15}$
 (C) $\frac{7}{15}$ (D) $\frac{8}{15}$
3. **Statement I:** In a pie chart, the sum of all central angles of the sectors is always
Statement II: The total value of the data in a pie chart is proportional to 360° .
 (A) Both Statement I and Statement II are true
 (B) Both Statement I and Statement II are false
 (C) Statement I is true and Statement II is false
 (D) Statement I is false and Statement II is true
4. The area of a rectangle is $x^2 + 5x + 6$. If its length is $x + 3$, what is its width?
 (A) $x + 2$ (B) $x + 1$
 (C) $x - 2$ (D) $x - 1$
5. A quadrilateral $ABCD$ has $\angle A = 70^\circ$, $\angle B = 110^\circ$, $\angle C = 70^\circ$, and $\angle D = 110^\circ$. What type of quadrilateral is it?
 (A) Kite (B) Trapezium
 (C) Parallelogram (D) Rhombus
6. A triangular garden has sides of lengths 13 m, 14 m, and 15 m. Using Heron's formula, find the cost of fencing it at ₹25 per meter.
 (A) ₹525 (B) ₹1050
 (C) ₹1575 (D) ₹2100
7. The decimal expansion of $\frac{7}{125}$ terminates after how many decimal places?
 (A) 2 (B) 3
 (C) 4 (D) 5
8. A polynomial $3x^2 + px + q$ has zeroes at $x = 2$ and $x = -3$. Then $p + q =$
 (A) -7 (B) -15
 (C) 13 (D) 7

9. In a trapezium, the parallel sides are in the ratio 3:5 and the distance between them is 12 cm. If the area is 384 cm^2 , what is the length of the longer parallel side?
 (A) 20 cm (B) 25 cm
 (C) 30 cm (D) 40 cm
10. The exterior angles of a triangle are in the ratio 2:3:4. What is the smallest interior angle?
 (A) 20° (B) 60°
 (C) 80° (D) 100°
11. If $\sqrt{0.0064} = 0.08$, then $\sqrt{64} + \sqrt{0.64}$ equals:
 (A) 8.8 (B) 8.08
 (C) 10.8 (D) 12.8
12. In a city park, two straight walking paths intersect. If one of the four angles formed at their intersection measures 52° , what is the measure of the angle vertically opposite to it?
 (A) 38° (B) 52°
 (C) 128° (D) 90°
13. In a triangular park, two sides are 12 m and 16 m. If the perimeter is 42 m, what is the length of the third side?
 (A) 10 m (B) 14 m
 (C) 26 m (D) 30 m
14. If $P(x) = 2x^3 - 4x^2 + 3x - 1$ and $Q(x) = x^2 - x + 2$, what is the degree of $P(x) \times Q(x)$?
 (A) 3 (B) 4
 (C) 5 (D) 6
15. The sides of a triangle are in the ratio 3:4:5 and its perimeter is 144 cm. Find its area using Heron's formula.
 (A) 864 cm^2 (B) 432 cm^2
 (C) 216 cm^2 (D) 1728 cm^2
16. The volume of a cuboid is $x^3 - 6x^2 + 11x - 6$. If one dimension is $x - 1$, what could be the other two dimensions?
 (A) $x - 2$ and $x - 3$
 (B) $x + 2$ and $x + 3$
 (C) $x - 2$ and $x + 3$
 (D) $x + 1$ and $x - 6$
17. A quadrilateral has three angles measuring 85° , 95° , and 110° . What is the measure of the fourth angle?
 (A) 70° (B) 80°
 (C) 90° (D) 100°
18. Which of these is an irrational number between $\frac{3}{7}$ and $\frac{5}{7}$?
 (A) 0.428571 (B) 0.5
 (C) $\sqrt{0.25}$ (D) 0.515115111...
19. Which of the following is not correctly matched?
- | Quadrilateral Type | Property |
|--------------------|--|
| (A) Rectangle | Diagonals are equal and bisect each other. |
| (B) Rhombus | Diagonals are equal and perpendicular. |
| (C) Square | Diagonals are equal, perpendicular, and bisect each other. |
| (D) Parallelogram | Diagonals bisect each other. |

20. A survey was conducted in a colony to find the number of pets per household. The data collected was: 2, 1, 0, 3, 2, 2, 1, 4, 0, 2, 1, 3. What is the modal number of pets per household?

- (A) 1 (B) 2
(C) 3 (D) 4

21. In a factory, the weekly production (in units) for five weeks was recorded as: 120, 135, 125, 140, 130. If the mean production is 130 units, what will be the production in the sixth week to maintain the same mean?

- (A) 120 (B) 130
(C) 140 (D) 150

22. The pictograph shows the number of trees planted by students of four classes in a school. Each  represents 10 trees



What is the average number of trees planted per class?

- (A) 35 (B) 40
(C) 45 (D) 50

23. In a geometry model, three rays OA, OB, OC originate from point O with $\angle AOB = 50^\circ$ and $\angle BOC = 70^\circ$. If OA and OC are on opposite sides of OB, what is $\angle AOC$?

- (A) 20° (B) 120°
(C) 60° (D) 130°

24. A triangular flag has sides measuring 7 cm, 24 cm, and 25 cm. What type of triangle is it?

- (A) Acute-angled
(B) Obtuse-angled
(C) Right-angled
(D) Equilateral

25. The heights (in cm) of 8 students are: 150, 152, 148, 155, 160, 148, 153, 150. What is the frequency of the height 148 cm?

- (A) 1 (B) 2
(C) 3 (D) 4

26. The following frequency table shows the marks obtained by students in a class test:

Marks	0–10	10–20	20–30	30–40
Frequency	5	8	12	5

How many students scored less than 30 marks?

- (A) 13 (B) 20
(C) 25 (D) 30

27. A solid is composed of a cylinder with a cone on top, both having the same radius of 7 cm. The height of the cylinder is 20 cm and the slant height of the cone is 25 cm. What is the total surface area of the solid?

- (A) 1584 cm^2 (B) 1672 cm^2
(C) 1848 cm^2 (D) 2024 cm^2

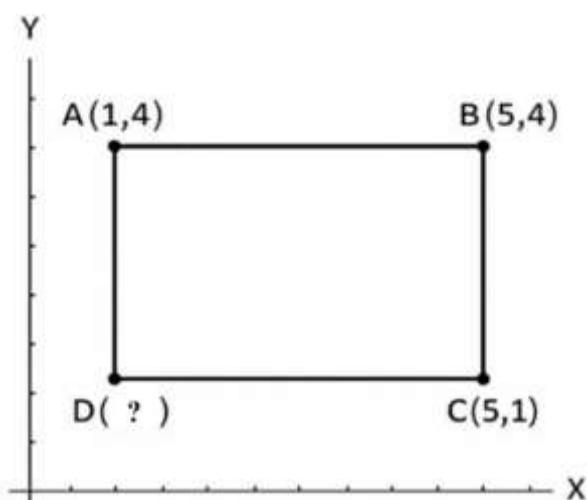
28. A metal cube of side 14 cm is melted and recast into small cubes of side 2 cm each. How many such small cubes can be made?

- (A) 343 (B) 294
(C) 196 (D) 144

29. The lateral surface area of a cube is 256 cm^2 . What is its volume?

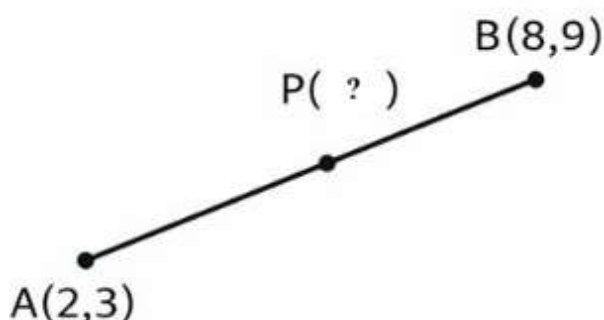
- (A) 64 cm^3 (B) 216 cm^3
(C) 512 cm^3 (D) 1024 cm^3

30. In the diagram below, ABCD is a rectangle. Coordinates of A and C are given. What are the coordinates of point D?



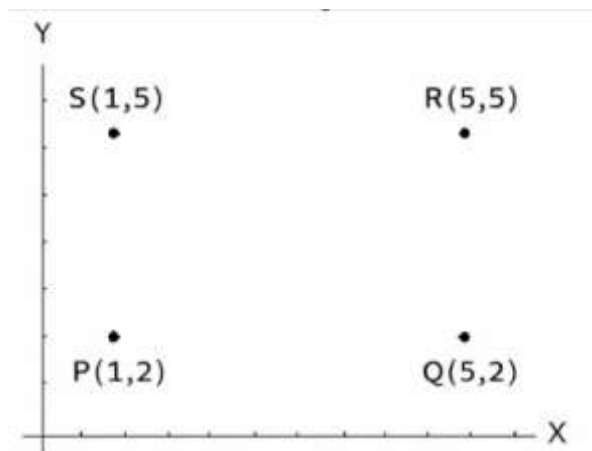
- (A) (1, 1) (B) (1, 5)
(C) (5, 1) (D) (4, 1)

31. The point P divides the line segment AB in the ratio 2:1 internally. If $A = (2, 3)$ and $B = (8, 9)$, which diagram correctly shows P?



- (A) $P = (4, 5)$ (B) $P = (5, 6)$
(C) $P = (6, 7)$ (D) $P = (7, 8)$

32. The quadrilateral PQRS is plotted below. Which type of quadrilateral is it?



- (A) Square (B) Rectangle
(C) Rhombus (D) Trapezium

33. A surveyor marks three points P, Q, R on a map such that Q lies exactly between P and R. If $PQ = 8 \text{ km}$ and $QR = 15 \text{ km}$, what is PR?

- (A) 7 km (B) 23 km
(C) 17 km (D) 10 km

34. A solar panel is installed such that its surface is parallel to the ground. If sunlight falls at an angle of 34° to the ground, what is the angle between the sunlight and the panel surface?

- (A) 34°
(B) 56°
(C) 146°
(D) 124°

35. A hemispherical bowl of radius 14 cm is completely filled with water. This water is poured into a cylindrical vessel of radius 7 cm. What will be the height of water in the cylinder?

- (A) 14 cm (B) $28/3 \text{ cm}$
(C) $112/3 \text{ cm}$ (D) 22 cm

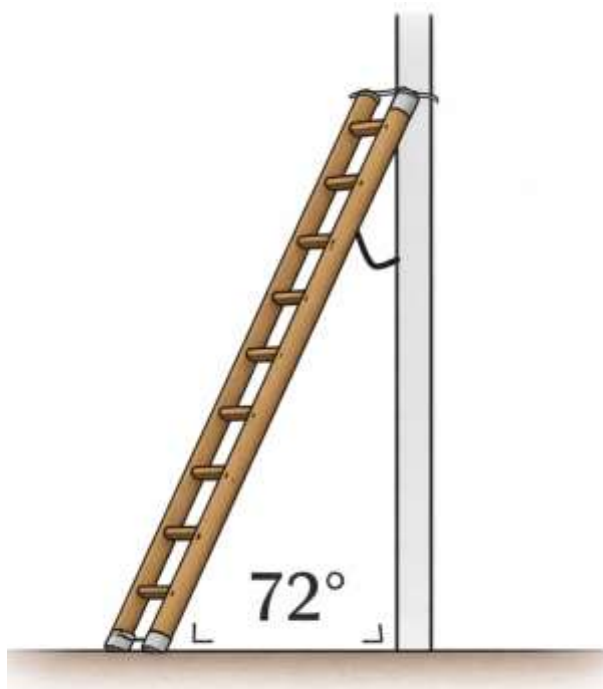
36. A cuboid has dimensions $12 \text{ cm} \times 10 \text{ cm} \times 8 \text{ cm}$. If all dimensions are increased by 50%, what is the percentage increase in its volume?
 (A) 50% (B) 125%
 (C) 237.5% (D) 337.5%
37. The sum of digits of a two-digit number is 9. If the number is reversed, it decreases by 45. The original number is:
 (A) 54 (B) 63
 (C) 72 (D) 81
38. A circular race track has an inner radius of 28 m and outer radius of 35 m. What is the width of the track?
 (A) 7 m (B) 14 m
 (C) 21 m (D) 63 m
39. The minute hand of a clock is 14 cm long. How much distance does its tip cover between 3:15 PM and 3:45 PM?
 (A) 44 cm (B) 88 cm
 (C) 22 cm (D) 66 cm
40. The ratio of the radii of two cylinders is 2:3 and their heights are in the ratio 5:4. What is the ratio of their volumes?
 (A) 5:9 (B) 10:9
 (C) 8:15 (D) 20:27
41. The cost of 3 pens and 5 pencils is ₹70, while 5 pens and 3 pencils cost ₹74. What is the cost of one pen?
 (A) ₹10 (B) ₹12
 (C) ₹14 (D) ₹16
42. The below table shows the number of students participating in different sports in a school.

Sport	Number of Students
Cricket	20
Football	30
Basketball	15
Badminton	10

If 5 students shift from Cricket to Basketball and 5 students shift from Football to Badminton, then which sport will have the second highest number of students after these changes?

- (A) Cricket (B) Football
 (C) Basketball (D) Badminton
43. A kite has a perimeter of 70 cm. One of the adjacent equal pairs has sides of length 20 cm each. What are the lengths of the other two sides?
 (A) 20 cm and 20 cm
 (B) 10 cm and 10 cm
 (C) 15 cm and 15 cm
 (D) 25 cm and 25 cm
44. The solution to $\frac{3x-2}{4} - \frac{2x+5}{3} = \frac{2}{3}$ is:
 (A) $x = 8$
 (B) $x = 30$
 (C) $x = 12$
 (D) $x = 34$
45. Three friends are standing in a line at equal distances. If the distance between the first and third is 24 meters, how far is the second from the first?
 (A) 8 m (B) 12 m
 (C) 16 m (D) 24 m

- 46.** A train covers a certain distance in 4 hours. If its speed is reduced by 10 km/h, it takes 5 hours. The original speed of the train is:
- (A) 40 km/h (B) 50 km/h
(C) 60 km/h (D) 70 km/h
- 47.** A quadrilateral is formed by joining the midpoints of the sides of a rectangle of sides **6 cm** and **8 cm**. What is the perimeter of this new quadrilateral?
- (A) 20 cm (B) 28 cm
(C) 40 cm (D) 24 cm
- 48.** A ladder leaning against a wall makes an angle of 72° with the ground. If the wall is perfectly vertical, what angle does the ladder make with the wall?



- (A) 18° (B) 72°
(C) 108° (D) 90°
- 49.** The area of a circle is equal to the sum of areas of two circles of radii 6 cm and 8 cm. What is the radius of the larger circle?
- (A) 10 cm (B) 14 cm
(C) 12 cm (D) 16 cm
- 50.** A student pumps air into a spherical balloon. The radius of the balloon increases from 7cm to 14cm. What is the ratio of the surface area of the balloon in the first case to that in the second case?
- (A) 1:4 (B) 1:2
(C) 4:1 (D) 1:8

